

Javier Rando

AI Safety Researcher. Figuring out what can go wrong when we deploy AI in real-world applications.

Email: javier.rando@ai.ethz.ch

Homepage: <https://javirando.com>

Education

PhD in Computer Science | ETH Zürich | Sep. 2023 - ongoing

Topic: **Real-world security and safety of AI systems.**

Advised by Prof. Florian Tramèr and Prof. Mrinmaya Sachan.

Awarded with the ETH AI Center Doctoral Fellowship 🏆.

MSc in Computer Science | ETH Zürich | 2021 – 2023

GPA: 5.64/6 with research projects on Language Models for password modeling (Prof. Fernando Perez-Cruz) and poisoning RLHF (Prof. Florian Tramèr).

Funded by an Open Philanthropy grant 🏆.

BSc in Mathematical Engineering in Data Science | Pompeu Fabra University | 2017 – 2021

First in class. Best Academic Record (9.1/10) 🏆.

Year abroad in the **Technical University of Munich (TUM)**.

Research and Industry Experience

Meta | Research Scientist Intern | Summer 2024

GenAI Safety & Trust Team.

OpenAI Red Teaming Network | Independent Contractor | 2024 – ongoing

Red teaming to assess the risks and safety profile of OpenAI models and systems.

New York University | Visiting Researcher | 2022 – 2023

Visiting Researcher in the CILVR with **Prof. He He**. Funded by a Long-Term Future Fund grant.

Topic: Language Models truthfulness.

EXPAI | Co-Founder and CTO | 2020 – 2023

Explainable AI startup. Provides a tool to understand AI predictions to boost efficiency and trust.

In charge of corporate strategy, innovation and product development.

Telefonica | Data Science and Engineering Intern | 2020

Pompeu Fabra University | Research Assistant | 2019

Supervised by Prof. Carlos Castillo and Dr. Valerio Lorini.

Project funded with a Maria de Maetzu award for talented undergraduate students 🏆.

Publication Uneven Coverage of Natural Disasters in Wikipedia: The Case of Floods published at IS-CRAM 2020.

Publications

Pre-prints

2024. *Persistent Pre-Training Poisoning of LLMs*. Yiming Zhang*, Javier Rando*, Jianfeng Chi, Eric Michael Smith, Nicholas Carlini, Florian Tramèr, Daphne Ippolito. *Under review at ICLR*.
2024. *Gradient-based Jailbreak Images for Multimodal Fusion Models*. Javier Rando, Hannah Korevaar, Erik Brinkman, Ivan Evtimov, Florian Tramèr. *Under review at ICLR*.
2024. *AutoAdvExBench: Benchmarking Autonomous Exploitation of Adversarial Example Defenses*. Nicholas Carlini, Edoardo Debenedetti, Milad Nasr, Javier Rando, Florian Tramèr. *Under review at ICLR*.
2024. *Measuring Non-Adversarial Reproduction of Training Data in Large Language Models*. Michael Aerni, Javier Rando, Edoardo Debenedetti, Nicholas Carlini, Daphne Ippolito, Florian Tramèr. *Under review at ICLR*.
2024. *An Adversarial Perspective on Machine Unlearning for AI Safety*. Jakub ucki, Boyi Wei, Yangsibo Huang, Peter Henderson, Florian Tramèr, Javier Rando. *Under review at NeurIPS Workshops*.
2024. *Scalable Extraction of Training Data from Aligned, Production Language Models*. Milad Nasr*, Javier Rando*, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, et al. *Under review at ICLR*.
2024. *Competition Report: Finding Universal Jailbreak Backdoors in Aligned LLMs*. Javier Rando, Francesco Croce, Krystof Mitka, Stepan Shabalin, Maksym Andriushchenko, Nicolas Flammarion, and Florian Tramèr. *Under review at NeurIPS*.

Proceedings

2024. *Dataset and Lessons Learned from the 2024 SaTML LLM Capture-the-Flag Competition*. Edoardo Debenedetti*, Javier Rando*, Daniel Paleka*, Silaghi Fineas Florin, et al. *NeurIPS Datasets and Benchmarks*. 🏆 **Spotlight** 🏆
2024. *Foundational challenges in assuring alignment and safety of large language models*. Usman Anwar, Abulhair Saparov, Javier Rando, Daniel Paleka, Miles Turpin, et al. TMLR.
2024. *Universal Jailbreak Backdoors from Poisoned Human Feedback*. Javier Rando, Florian Tramèr. ICLR. 🏆 **2nd Prize at the AI Safety Prize Competition**. 🏆
2024. *Personas as a way to Model Truthfulness in Language Models*. Nitish Joshi*, Javier Rando*, Abulhair Saparov, Najoung Kim, He He. EMNLP 2024.
2024. *Attributions toward artificial agents in a modified Moral Turing Test*. Eyal Aharoni, Sharlene Fernandes, Daniel J. Brady, Caelan Alexander, Michael Criner, Kara Queen, Javier Rando, Eddy Nahmias, and Victor Crespo. Scientific Reports.
2023. *Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback*. Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, J  r  my Scheurer, Javier Rando, et al. Transactions on Machine Learning Research (TMLR).
2023. *PassGPT: Password Modeling and (Guided) Generation with Large Language Models*. Javier Rando, Fernando Perez-Cruz, Briland Hitaj. ESORICS.
2022. *"That Is a Suspicious Reaction!": Interpreting Logits Variation to Detect NLP Adversarial Attacks*. Edoardo Mosca, Shreyash Agarwal, Javier Rando, Georg Groh. ACL.
2020. *Uneven Coverage of Natural Disasters in Wikipedia: The Case of Floods*. Valerio Lorini, Javier Rando, Diego Saez-Trumper, Carlos Castillo. ISCRAM 2020 Conference Proceedings.

Workshops

2024. *Adversarial Perturbations Cannot Reliably Protect Artists From Generative AI*. Robert Hönig, Javier Rando, Nicholas Carlini, and Florian Tramèr. GenLaw Workshop @ ICML. 🏆 **Spotlight** 🏆.

2023. *Jailbreaking Language Models at Scale via Persona Modulation*. Rusheb Shah, Quentin Feuillade-Montixi, Soroush Pour, Arush Tagade, Javier Rando. SoLaR Workshop @ NeurIPS.

2022. *Red-teaming the Stable Diffusion Safety Filter*. Javier Rando, Daniel Paleka, David Lindner, Lennart Heim, Florian Tramèr. NeurIPS ML Safety Workshop. 🏆 **Best Paper Award** 🏆.

2022. *How is Real-World Gender Bias Reflected in Language Models?* Javier Rando, Alexander Theus, Rita Sevastjanova, Menna El-Assady. IEEE VIS VISxAI Workshop.

2022. *Exploring Adversarial Attacks and Defenses in Vision Transformers trained with DINO*. Javier Rando, Thomas Baumann, Nasib Naimi, Max Mathys. ICML Adversarial ML Frontiers Workshop.

Teaching

Large Language Models (2024). Teaching assistant. *ETH Zurich*.

Information Security Lab (2024). Teaching assistant. *ETH Zurich*.

NYU AI School (2023). Teaching assistant. *New York University*.

Service

Organizing committee

2023 **Competition organizer at SaTML 2023**. *Large Language Models Capture the Flag* and *Find the Trojan: Universal Backdoor Detection in Aligned Large Language Models*

Peer reviewer

2024 **ICLR**. Main conference track.

2024 **ICML**. Main conference track.

2023, 2024 **Socially Responsible Language Modelling Research (SoLaR) Workshop** at NeurIPS.

Student Supervision

Robert Hönig (MSc Student)

Jakub Lucki (MSc Student)

Thomas Baumann (MSc Student)

Awards and Grants

2023 **ETH AI Center Doctoral Fellowship.**

2022 **Best Paper Award** at *NeurIPS ML Safety Workshop*.

2022 **Runner up** at *Junction 2022 Hackathon Helsinki*. Challenge: *Crypto Trading Helper*.

2022 **Research grant** (\$30K) to join NYU as Visiting Researcher. *Long-Term Future Fund*.

2021 **Early career scholarship** for MSc funding (\$30K). *Open Philanthropy Project*.

2021 **Best Academic Record Award**. BSc Mathematical Engineering in Data Science (9.1/10). *Pompeu Fabra University*.

2019 **Collaboration scholarship at units of excellence "María De Maeztu"**. Research assistant funding at Pompeu Fabra University. *Spanish Science Ministry*.

2020 **TALENTUM internship Telefonica**. Internship to join an applied research team. *Telefonica*.

Invited Talks

2024 **Princeton PLI Seminar**. Adversarial perturbations cannot reliably protect artists from Gen. AI.

2021 **Pompeu Fabra University**. Guest lectures (4h). Visual Analytics undergraduate course.

2021 **Pompeu Fabra University**. Guest lecture (2h). Entrepreneurship undergraduate course.

Last updated: October 2, 2024